Kernix 0.11.1 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

35 workloads

119 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



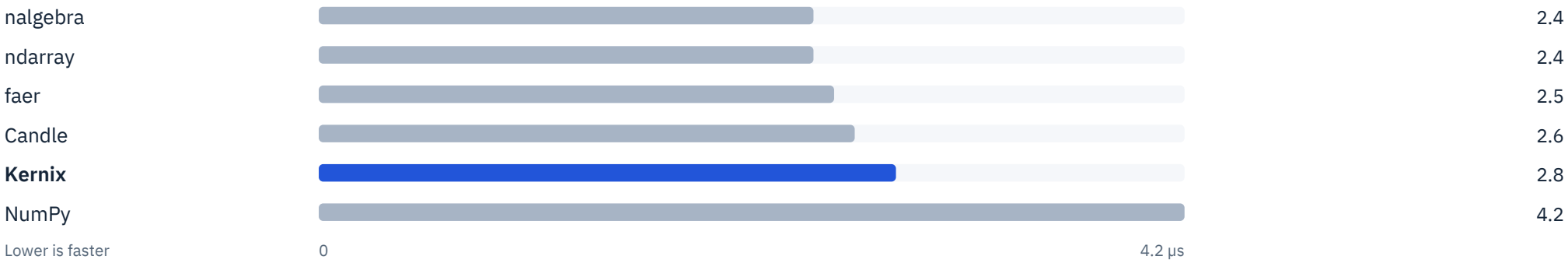
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



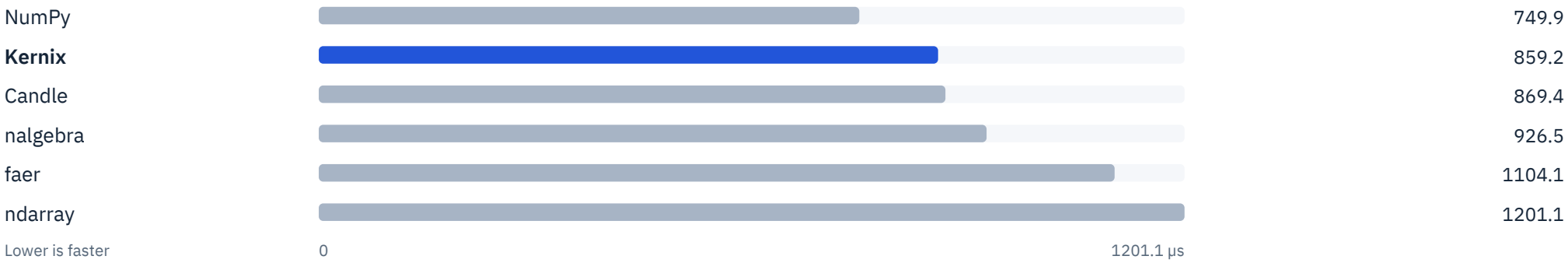
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



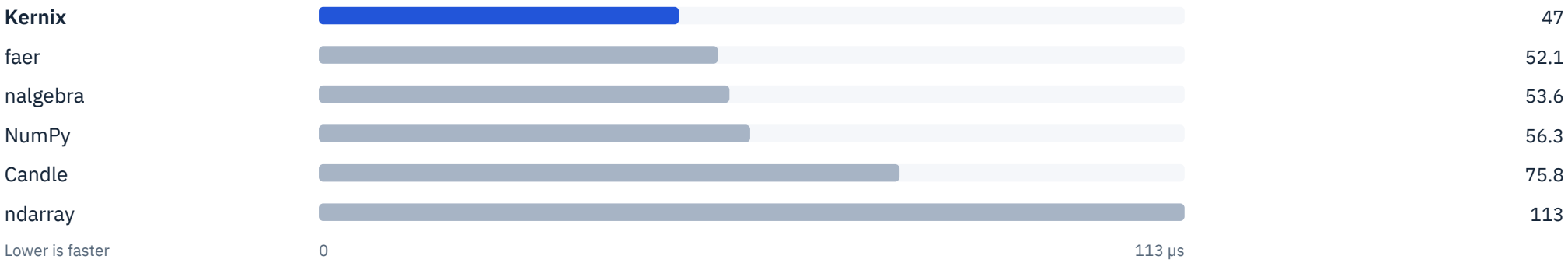
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

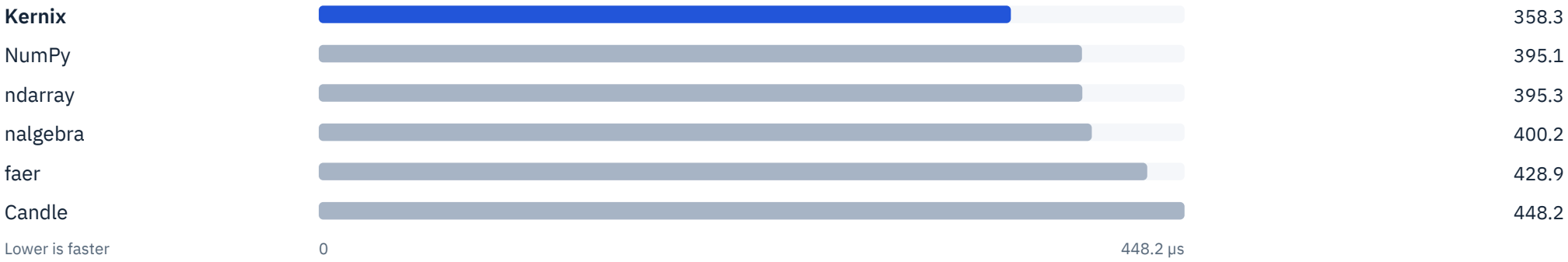
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



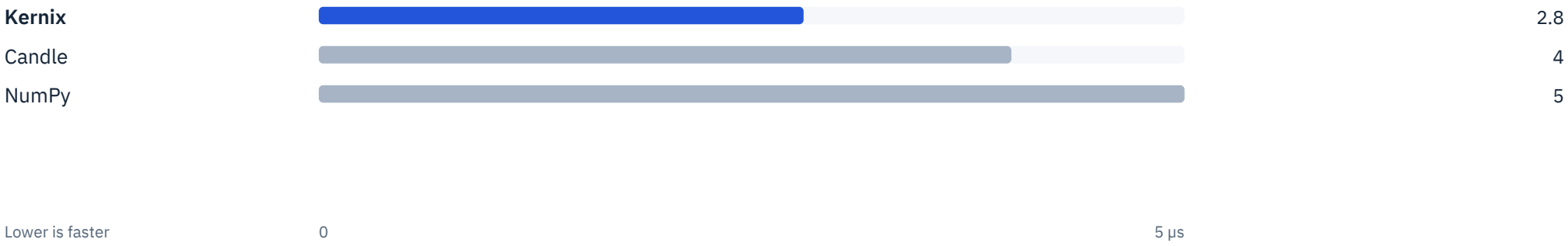
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

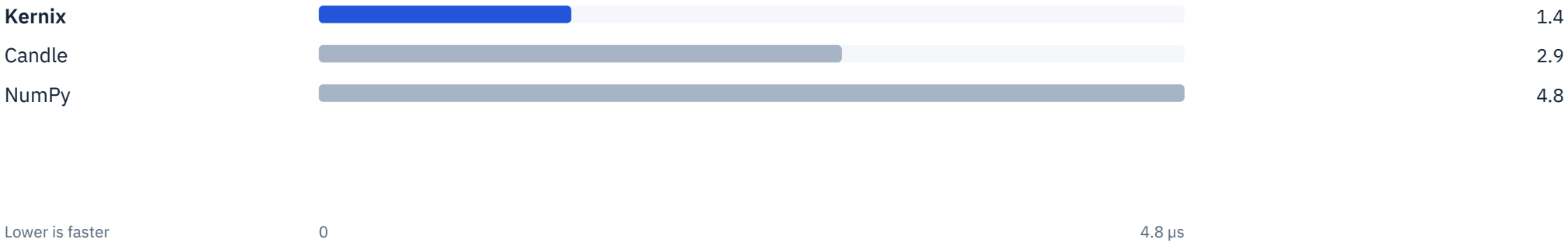
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

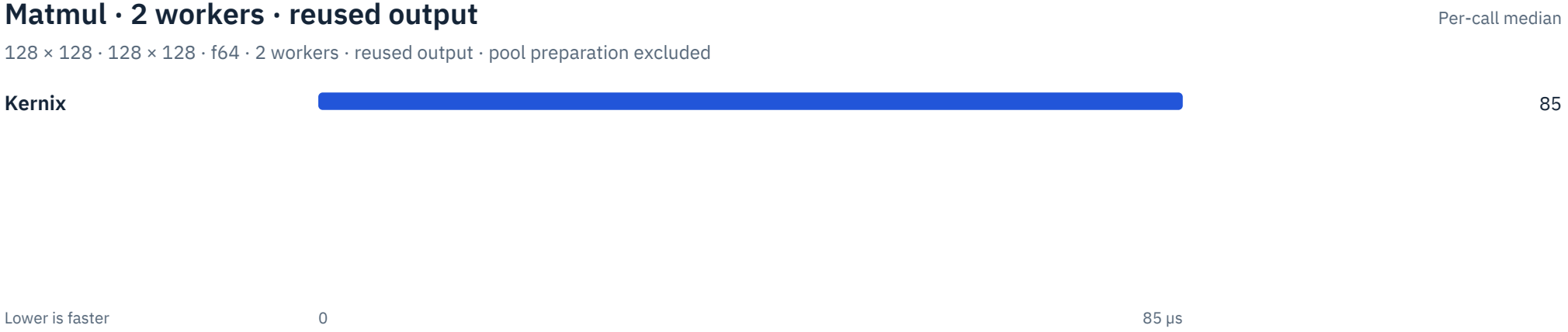
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded

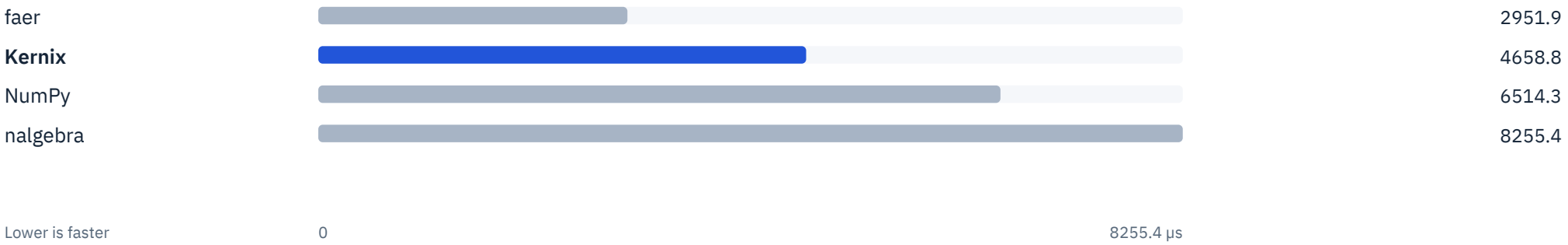


Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

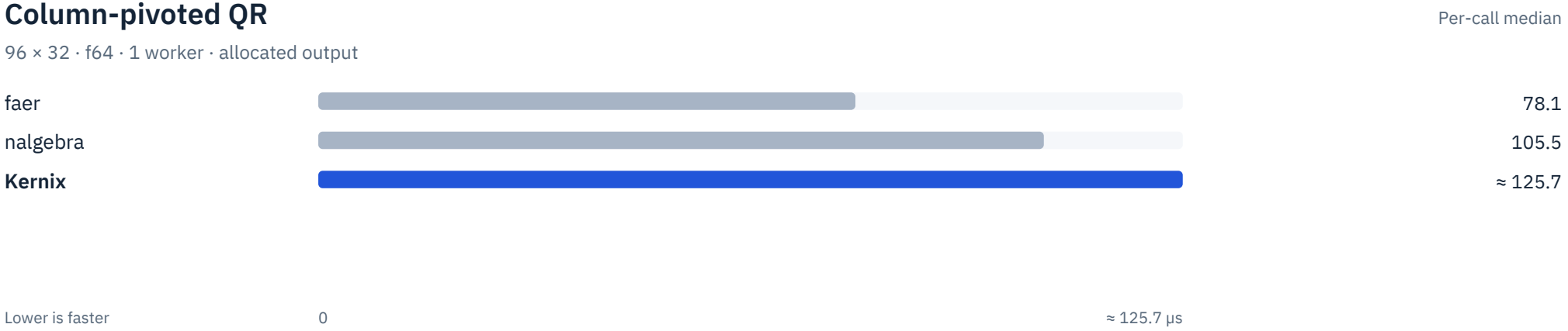
Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

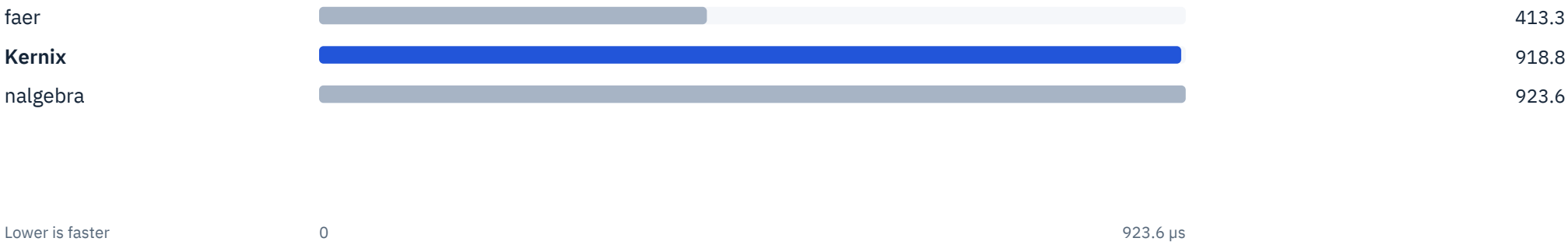


Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



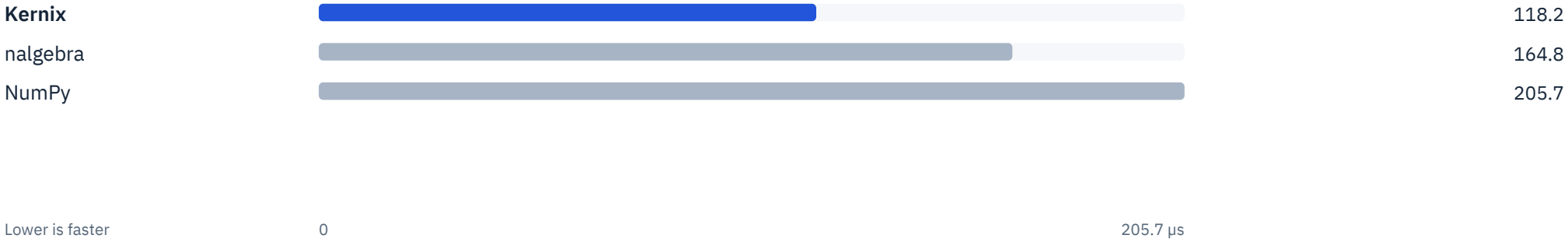
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output

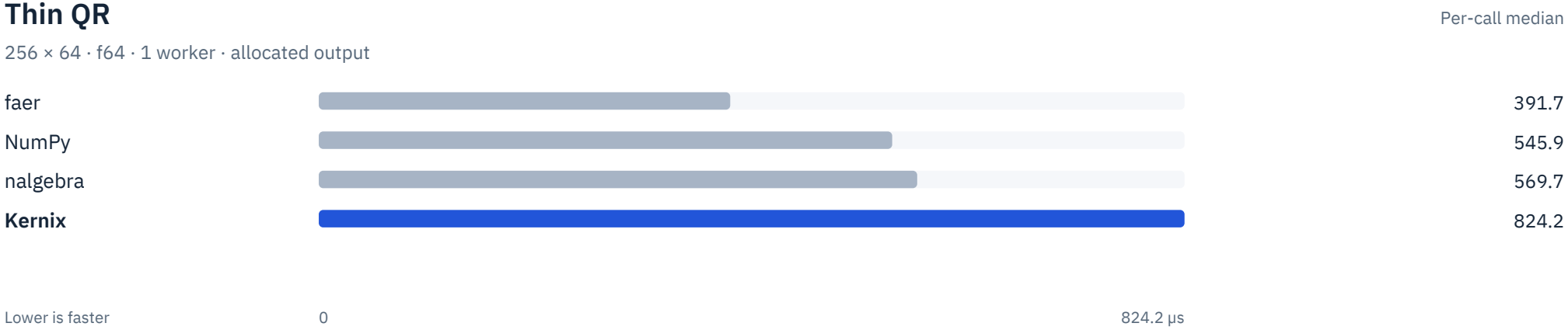


Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

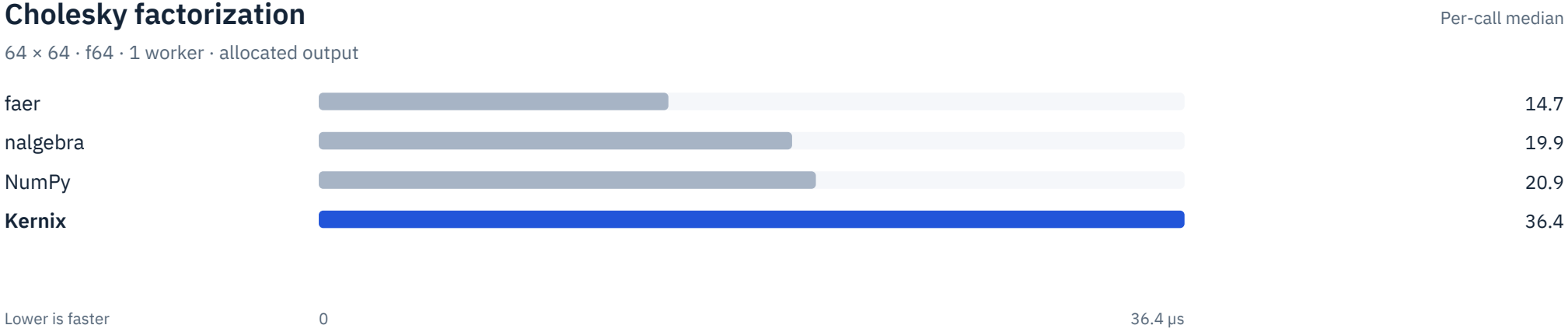
Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



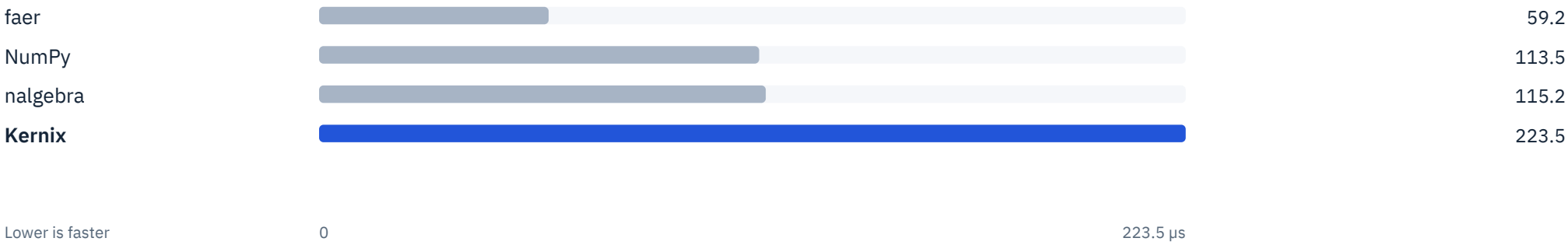
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

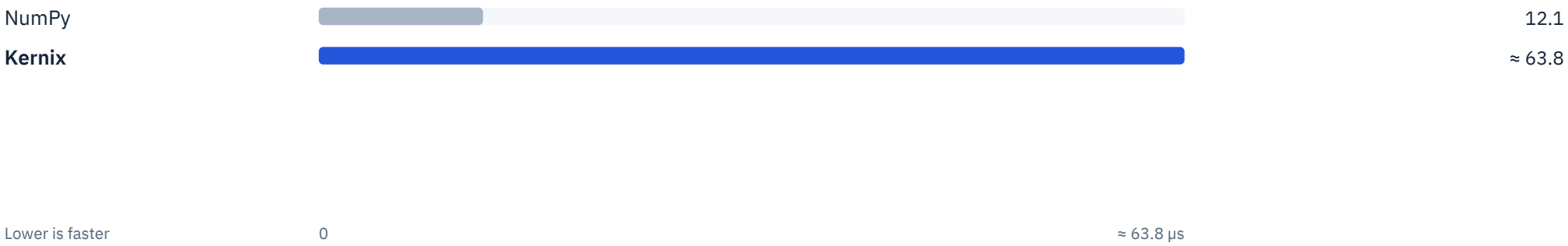


Kernix 0.11.1 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



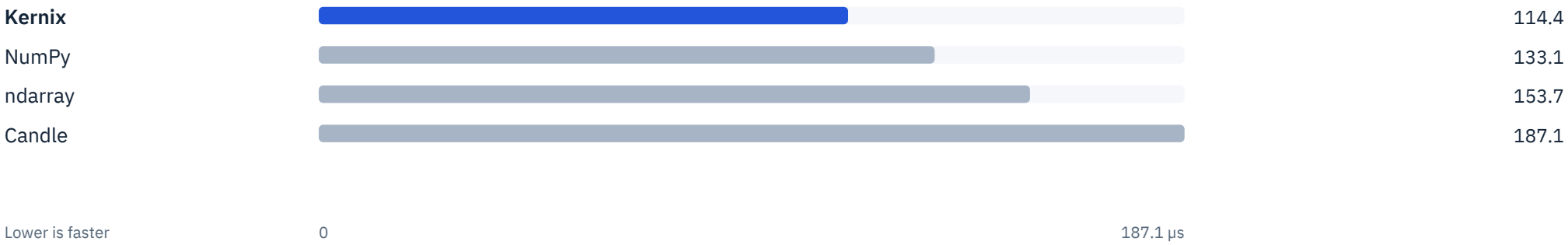
Kernix 0.11.1 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1	1 – 1

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0031813361611877	0.9947478991596639 – 1.4287222808870117

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0021276595744681	0.9978768577494692 – 1.004255319148936

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0005473453749316	0.9873041599135602 – 1.155377946356001

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0005652911249294	0.9435931601154786 – 1.1982515510434293

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9245283018867925	0.9074074074074074 – 0.9433962264150944

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0043859649122806	0.6431924882629108 – 1.1081277213352685

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0027510316368637	0.9773936170212766 – 1.0097087378640777

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7142857142857143	0.6944444444444444 – 0.7222222222222222

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.6666666666666666	0.6666666666666666 – 0.75

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0004301075268818	0.9575835475578406 – 1.0379633765073693

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9966810966810967	0.9867199771526489 – 1.0048934945308001

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0021162437741817	0.9961633557848069 – 1.0049226291120006

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9976331360946745	0.9316427783902976 – 1.005966587112172

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0054669703872436	0.9654383154039826 – 1.0339377033937704

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.99673735725938	0.9522240527182867 – 1.0662020905923344

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	104300	104.3
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	141800	141.8
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	56400	56.4
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	71200	71.2
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2600	2.6
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4200	4.2
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2800	2.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	61500.00000000001	61.50000000000001
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	16000	16
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14200	14.2
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14600	14.6
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	58500	58.5
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13500	13.5
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	101200	101.2
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	114400	114.4
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	155200	155.2
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	110200	110.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	116000	116
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	137300	137.3
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	749900	749.9
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	869400	869.4
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	926500	926.5
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1201100	1201.1
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1104100	1104.1
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	859200	859.2
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	29000	29
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	10500	10.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7000	7
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	14000	14
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7400	7.4
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7800	7.8
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53600	53.6
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	52100	52.1
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	75800	75.8
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	113000	113
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	47000	47
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	56300	56.3

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	400200	400.2
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	395100	395.1
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	395300	395.3
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	428900	428.9
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	358300	358.3
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	448200	448.2
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4000	4
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	5000	5
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2800	2.8
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2900	2.9

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	4800	4.8
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1400	1.4
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	32700.000000000004	32.700000000000004
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	85000	85
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	501600.00000000006	501.60000000000006
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	58600	58.6
Cholesky factorization / nalgebra 512 × 512 · f64	8255400	8255.4
Cholesky factorization / NumPy 512 × 512 · f64	6514300	6514.3
Cholesky factorization / faer 512 × 512 · f64	2951900	2951.9
Cholesky factorization / Kernix 512 × 512 · f64	4658800	4658.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	78100	78.1
Column-pivoted QR / nalgebra 96 × 32 · f64	105500	105.5
Column-pivoted QR / Kernix 96 × 32 · f64	125699.99999999999	125.69999999999999
Column-pivoted QR / nalgebra 256 × 64 · f64	923600	923.6
Column-pivoted QR / faer 256 × 64 · f64	413300	413.3
Column-pivoted QR / Kernix 256 × 64 · f64	918800	918.8
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	105300	105.3
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	114600	114.6
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	94900	94.9
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	205700	205.7

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	118200	118.2
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	164800	164.8
Thin QR / faer 96 × 32 · f64	84300	84.3
Thin QR / NumPy 96 × 32 · f64	76800	76.8
Thin QR / Kernix 96 × 32 · f64	100600	100.6
Thin QR / nalgebra 96 × 32 · f64	78300	78.3
Thin QR / faer 256 × 64 · f64	391700	391.7
Thin QR / nalgebra 256 × 64 · f64	569700	569.7
Thin QR / Kernix 256 × 64 · f64	824200	824.2
Thin QR / NumPy 256 × 64 · f64	545900	545.9

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	36400	36.4
Cholesky factorization / faer 64 × 64 · f64	14700	14.7
Cholesky factorization / NumPy 64 × 64 · f64	20900	20.9
Cholesky factorization / nalgebra 64 × 64 · f64	19900	19.9
Cholesky factorization / NumPy 128 × 128 · f64	113500	113.5
Cholesky factorization / nalgebra 128 × 128 · f64	115200	115.2
Cholesky factorization / faer 128 × 128 · f64	59200	59.2
Cholesky factorization / Kernix 128 × 128 · f64	223500	223.5
Cholesky factorization / NumPy 256 × 256 · f64	1002000	1002
Cholesky factorization / faer 256 × 256 · f64	316800	316.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	832800	832.8
Cholesky factorization / Kernix 256 × 256 · f64	703500	703.5
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	50700	50.7
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	9800	9.8
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	10200	10.2
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4100	4.1
Model inference / Candle batched normalized/sliced MLP · f64	8400	8.4
Model inference / Kernix batched normalized/sliced MLP · f64	3500	3.5
Model recording + forward / Kernix batched normalized/sliced MLP · f64	5000	5
Model recording + forward / Candle batched normalized/sliced MLP · f64	10100	10.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	5800	5.8
Model backward / Candle prepared graph → fresh gradients · f64	32700.000000000004	32.700000000000004
Parameter update / Kernix transactional SGD, fixed gradients · f64	1000	1
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63800.00000000001	63.80000000000001
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	12100	12.1
Row softmax / Candle 256 × 64 · f64	187100	187.1
Row softmax / ndarray 256 × 64 · f64	153700	153.7
Row softmax / Kernix 256 × 64 · f64	114400	114.4
Row softmax / NumPy 256 × 64 · f64	133100	133.1
Create zeros / faer 64 × 64 · f32	500	0.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	500	0.5
Create zeros / NumPy 64 × 64 · f32	1300	1.3
Create zeros / Kernix 64 × 64 · f32	400	0.4
Create zeros / ndarray 64 × 64 · f32	300	0.3
Create ones / Candle 64 × 64 · f32	700	0.7
Create ones / ndarray 64 × 64 · f32	400	0.4
Create ones / NumPy 64 × 64 · f32	2100	2.1
Create ones / Kernix 64 × 64 · f32	600	0.6
Create ones / faer 64 × 64 · f32	500	0.5